Course Code PHY1004	Fundamentals of Space Physics	Course Type	LTP
		Credits	4

Course Objectives:

To understand fundamental physical aspects of some components of the space science explaining the basic processes in the interplanetary medium, in near-Earth space, and down into the Earth's atmosphere.

Course Outcomes:

CO1. Understanding of the mathematical formulations and development of problem solving approach with reference to Space Science.
CO2. Development of the reasoning associated with the concepts of reference frame, time and event which form the framework of the spacecraft kinematics.
CO3. Understanding of the fundamental physical processes in the Earth's and nearby atmosphere and/or space.
CO4. Understanding of the fundamental constructional and operational aspects involved in Spacecraft Technology.

Mod. No.	Module Description	Lect. Hrs	CO
1	**Mathematical Foundation** Coordinate Systems: Cartesian, Cylindrical and Spherical Polar; Vector Calculus: Gradient, Divergence and Curl and their Physical Significance and Applications in Nature; Line, Surface, and Volume integrals, Gauss's Divergence theorem and Stokes' Theorem and Their Applications. Dirac Delta function and Its Application and Physical Significance.	6	CO1
2	**Reference Frames and Time** Reference Frames, Ecliptic and Celestial Equatorial Planes. Reference Frames and Coordinate Axes (Earth-fixed geocentric equatorial frame, Topocentric frame, Geocentric equatorial reference frame, Heliocentric equatorial reference frame, Heliocentric ecliptic reference frame, Barycentric reference frame), Euler Angles and Transformations of Coordinates, Time Intervals and Epoch (International Atomic Time, Universal Time (UT) and Greenwich Mean Solar Time (GMST), Sidereal Time).	8	CO2
3	**Physics of Space Environment** Definition and Constraints, Brief History of Space Research, General Properties of Plasma; Definition, Plasma Production, Particle Interaction and Collective Effects, Basic Plasma Phenomena. Criteria for the definition of Plasma; Macroscopic Neutrality, Debye Shielding, Plasma Frequency.	6	CO3

	Occurrence of plasma in nature; The Sun and its Atmosphere, The Solar Wind, The **Planetary** magnetosphere (e.g., Earth's magnetosphere) and Van Allen Radiation Belts, The Ionosphere, Space Weather.		
4	**Physics of Space Plasma** Maxwell's Equations, Continuity Equation, Poynting Theorem, Wave solution of Maxwell Equations. Electromagnetic Waves, Particle Orbit Theory; Electromagnetic Fields, Particles in Constant External Fields, Nonuniform Magnetic Fields, Adiabatic Invariants. Kinetic Theory; Collisions, The Boltzmann Equation, Multispecies Gases. Fluid and MHD Theory; Moment Equations, Transport Equations for Multispecies Gasses.	**9**	**CO3**
5	**Fundamentals of Spacecraft Technology** Structure and Mechanisms; The Primary Structure of the Spacecraft, Secondary and Deployable Structures, Structural Analysis, Qualification of the Spacecraft Structure, Mechanisms Electrical Power Supply; Energy Generation, Power Sources, Electrical Power System Architectures, Solar Array, Energy Storage, Thermal Control; Basic Thermal Principles, Thermal Systems, Technical solutions of thermal issues, Examples of a Thermal Design, Thermal Control Systems	**9**	**CO4**
6	**Guest Lecture on Contemporary Topics**	**2**	
	Total Lecture Hours:	**40**	

(1) Activity based on divergence, curl, and gradients using simulation kit.
(2) Activity based on inertial and non-inertial frames.
(3) Solar-system models and simulations.
(4) Activity on relative frame of references attached to Earth, Sun, and other celestial objects.
(5) Activity on near-Earth plasma phenomena using CCMC models.
(6) Activities on realization of spacecraft payloads, deployable structures.

Mode of Teaching and Learning:
Flipped Class Room, Activity Based Teaching/Learning, Digital/Computer based models, wherever possible to augment lecture for practice/tutorial and minimum 2 hours lecture by industry experts on contemporary topics

Mode of Evaluation:
The assessment and evaluation components may consist of quizzes, assignments, tutorials and any other innovative assessment practices followed by faculty, in addition to the Continuous Assessment Tests and Term End Examination.

Text Book(s):

1.	Physics of the Space Environment, Tamas I. Gombosi, Cambridge University Press.
2.	Physics of the Earth's Space Environment; An Introduction, Gerd W Prolss, Springer.
3.	Elements of Space Technology. For Aerospace Engineers, Rudolf X. Meyer, Academic.
4.	Introduction to Electrodynamics, David J. Griffiths, Pearson, Cambridge University Press.
Reference Books:	
1.	Space Weather- Physics and Effects, Volker Bothmer and Ioannis A. Daglis, Springer
2.	The Physics of Fluids and Plasmas: An Introduction for Astrophysicists, Arnab Rai Choudhuri, Cambridge University Press.
3.	Handbook of Space Technology, Wilfried Ley,Klaus Wittmann,Willi Hallmann, John Wiley & Sons, Ltd, ISBN: 978-0-470-69739-9
Recommendation by the Board of Studies on	17/10/2020
Approval by Academic council on	20/10/2020
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